Glueing in Cubical Type Theory

Types, Terms, and the Univalence Axiom

Following the CCHM paper (Cohen, Coquand, Huber, Mörtberg, 2016)

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Contents

1 Motivation and Overview **2**

2 Prerequisites **2**
2.1 The interval and face formulas . . . 2
2.2 Composition . . . 2
2.3 Equivalences . . . 2

3 Glueing: Syntax and Inference Rules **3**
3.1 The three constructors . . . 3
3.2 Typing rules . . . 3
3.3 Equality rules . . . 3

4 Geometric Intuition **4**
4.1 The basic picture . . . 4
4.2 Path of types from an equivalence . . . 4
4.3 The general diagram (CCHM Figure 4 remark) . . . 5
4.4 Why `unglue` is the key map . . . 5

5 Composition for Glueing **5**
5.1 Setup and goal . . . 5
5.2 The algorithm (CCHM §6.2) . . . 5
5.3 Step-by-step intuition . . . 6

6 Composition for the Universe **6**
6.1 Constructing an equivalence from a line of types . . . 6
6.2 Composition for $\mathcal{U}$ . . . 6

7 From Equivalence to Path **6**

8 The Univalence Axiom as a Theorem **7**
8.1 The main theorem . . . 7
8.2 Contractibility and the univalence axiom . . . 7

9 Summary **8**

1 Motivation and Overview

The composition operation comp^i in CCHM expresses that *being extensible is preserved along paths*: given a partial element on face φ at i_0 that can be extended to a total element, it can also be extended at i_1. But what about *equivalences*? If a type T on face φ is related to a total type A by an equivalence $f : \mathsf{Equiv}\, T\, A$, can we "extend" terms across φ?

The **glueing construction** answers this. It introduces a new type former

$$\mathsf{Glue}[\varphi \mapsto (T, f)]\, A$$

that, on the face φ, looks exactly like T (glued in via f), and off φ looks like A. Glueing thus expresses:

Extensibility is invariant under equivalence.

From this single construction we can:

(1) Define **composition for the universe** $\mathcal{U}$ (which is otherwise not fibrant).

(2) Construct a **path from any equivalence** $f : \mathsf{Equiv}\, A\, B$ to a path $A = B$ in $\mathcal{U}$.

(3) Prove **Voevodsky's univalence axiom** as a *theorem* rather than an axiom.

2 Prerequisites

2.1 The interval and face formulas

The **interval** $\mathbb{I}$ has closed endpoints $i_0, i_1 : \mathbb{I}$ and generates **dimension variables** $i, j, k, \ldots$. A **face formula** $\varphi : \mathbb{F}$ is a formal conjunction/disjunction of atoms $i = i_0$ and $i = i_1$. We write $1_{\mathbb{F}}$ for the top formula (always true) and $0_{\mathbb{F}}$ for the bottom.

A **system** $[\varphi \mapsto u]$ denotes a partial element u defined on the face φ. Multiple faces $[\varphi_1 \mapsto u_1,\, \varphi_2 \mapsto u_2]$ are allowed provided $u_1 = u_2$ wherever $\varphi_1 \wedge \varphi_2$ holds.

2.2 Composition

The core operation (CCHM §4.3):

$$\frac{\Gamma \vdash \varphi : \mathbb{F} \quad \Gamma,\, i : \mathbb{I} \vdash A \quad \Gamma,\, \varphi,\, i : \mathbb{I} \vdash u : A \quad \Gamma \vdash a_0 : A(i_0)[\varphi \mapsto u(i_0)]}{\Gamma \vdash \mathsf{comp}^i\, A[\varphi \mapsto u]\, a_0 : A(i_1)[\varphi \mapsto u(i_1)]}$$

comp^i binds i in both A and u. The boundary condition is *built into the return type.*

2.3 Equivalences

Definition 2.1. *For a map $f : T \to A$ define*

$$\mathsf{isEquiv}\, T\, A\, f \;=\; (y : A) \to \mathsf{isContr}\big((x : T) \times \mathsf{Path}_A\, y\, (f\, x)\big)$$

An ***equivalence*** *is a pair $\mathsf{Equiv}\, T\, A = (f : T \to A) \times \mathsf{isEquiv}\, T\, A\, f$. When $f : \mathsf{Equiv}\, T\, A$ and $t : T$ we write $f\, t$ for $f.1\, t$.*

Lemma 2.2 (equiv, Lemma 7 of CCHM)**.** *If $\Gamma \vdash f : \mathsf{Equiv}\, T\, A$ then we have an operation*

$$\frac{\Gamma,\, \varphi \vdash t : T \quad \Gamma \vdash a : A \quad \Gamma,\, \varphi \vdash p : \mathsf{Path}_A\, a\, (f\, t)}{\Gamma \vdash \mathsf{equiv}\, f[\varphi \mapsto (t, p)]\, a : \big((x : T) \times \mathsf{Path}_A\, a\, (f\, x)\big)[\varphi \mapsto (t, p)]}$$

Given a partial lift (t, p) on φ and a base point $a : A$, we can produce a total element extending the partial one.

3 Glueing: Syntax and Inference Rules

3.1 The three constructors

CCHM introduces three new term formers (Figure 4 of the paper):

$$
\begin{array}{rcll}
t, u, A, B & ::= & \cdots \mid \mathsf{Glue}[\varphi \mapsto (T, f)]\, A & \text{Glue type} \\
& & \mid \mathsf{glue}[\varphi \mapsto t]\, u & \text{Glue term (introduction)} \\
& & \mid \mathsf{unglue}[\varphi \mapsto f]\, u & \text{Unglue term (elimination)}
\end{array}
$$

We may write $\mathsf{unglue}\, b$ as shorthand when f is clear from context.

3.2 Typing rules

Glue type formation

$$
\frac{\Gamma \vdash A \qquad \Gamma, \varphi \vdash T \qquad \Gamma, \varphi \vdash f : \mathsf{Equiv}\, T\, A}{\Gamma \vdash \mathsf{Glue}[\varphi \mapsto (T, f)]\, A}
$$

Unglue (elimination)

$$
\frac{\Gamma \vdash b : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A}{\Gamma \vdash \mathsf{unglue}[\varphi \mapsto f]\, b : A[\varphi \mapsto f\, b]}
$$

On the face φ, the result is exactly $f\, b$: unglue applies the equivalence map to recover an element of A.

Glue (introduction)

$$
\frac{\Gamma, \varphi \vdash f : \mathsf{Equiv}\, T\, A \qquad \Gamma, \varphi \vdash t : T \qquad \Gamma \vdash a : A[\varphi \mapsto f\, t]}{\Gamma \vdash \mathsf{glue}[\varphi \mapsto t]\, a : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A}
$$

3.3 Equality rules

$1_{\mathbb{F}}$-reduction (boundary collapse)

$$
\frac{\Gamma \vdash T \qquad \Gamma \vdash f : \mathsf{Equiv}\, T\, A}{\Gamma \vdash \mathsf{Glue}[1_{\mathbb{F}} \mapsto (T, f)]\, A = T} \qquad \frac{\Gamma \vdash t : T \qquad \Gamma \vdash f : \mathsf{Equiv}\, T\, A}{\Gamma \vdash \mathsf{glue}[1_{\mathbb{F}} \mapsto t]\, (f\, t) = t : T}
$$

When $\varphi = 1_{\mathbb{F}}$, the Glue type collapses to T and glue collapses to t. The gluing "disappears" when the face is everything.

η-rule for Glue

$$\frac{\Gamma \vdash b : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A}{\Gamma \vdash b = \mathsf{glue}[\varphi \mapsto b]\,(\mathsf{unglue}\, b) : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A}$$

Every element of a Glue type is definitionally equal to a glue of its own ungluing.

β-rule (unglue-glue)

$$\frac{\Gamma, \varphi \vdash f : \mathsf{Equiv}\, T\, A \qquad \Gamma, \varphi \vdash t : T \qquad \Gamma \vdash a : A[\varphi \mapsto f\, t]}{\Gamma \vdash \mathsf{unglue}\,(\mathsf{glue}[\varphi \mapsto t]\, a) = a : A}$$

Ungluing a glued element recovers the A-component.

Remark 3.1. *It follows from the rules that* $\Gamma \vdash b : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A$ *implies* $\Gamma, \varphi \vdash b : T$. *On the face* φ, *a Glue element* is *a* T*-element.*

4 Geometric Intuition

4.1 The basic picture

Think of A as a total type and T as a partial type living only on face φ. The equivalence $f : \mathsf{Equiv}\, T\, A$ says T and A are "the same" wherever φ holds.

$\mathsf{Glue}[\varphi \mapsto (T, f)]\, A$ is the type that:

- *outside* φ: looks exactly like A
- *on* φ: looks exactly like T
- the two views are coherently glued by f

4.2 Path of types from an equivalence

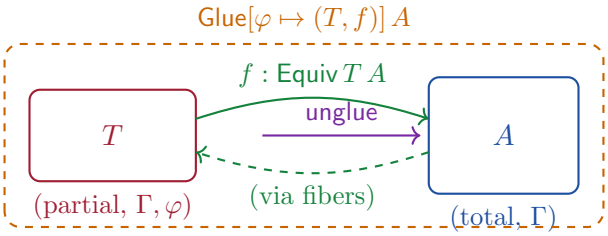
When $\varphi = (i = i_0) \vee (i = i_1)$, setting $T_0 = T(i_0)$, $T_1 = T(i_1)$, the picture (from CCHM §6.1) is:

$$\begin{tikzcd}
T_0 \arrow[r, "\sim", "f(i_0)"'] \arrow[d, dashed] & A(i_0) \arrow[d] \\
\mathsf{Glue}[\varphi \mapsto (T, f)]\, A \arrow[r, dashed, "\mathsf{unglue}"'] & A \\
T_1 \arrow[r, "f(i_1)", "\sim"'] \arrow[u, dashed] & A(i_1) \arrow[u]
\end{tikzcd}$$

The dashed vertical line is $\mathsf{Glue}[\varphi \mapsto (T, f)]\, A$: it bridges the two endpoints by gluing T_0 and T_1 onto the path A in the universe.

4.3 The general diagram (CCHM Figure 4 remark)

In general $\mathsf{Glue}[\varphi \mapsto (T, f)]\, A$ can be illustrated as:



$\mathsf{unglue} : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A \to A$ is the projection that *extends* f: it agrees with f wherever φ holds.

4.4 Why `unglue` is the key map

The theorem that unglue is itself an equivalence is precisely the univalence axiom. The intuition is:

- unglue is surjective: given $a : A$, use the equivalence f on the face φ to find the preimage $t : T$, then glue to get a total element of Glue.
- unglue is injective: the η-rule $b = \mathsf{glue}[\varphi \mapsto b](\mathsf{unglue}\, b)$ says every Glue element is fully determined by its unglue projection.

5 Composition for Glueing

5.1 Setup and goal

Assume $\Gamma,\, i : \mathbb{I} \vdash B = \mathsf{Glue}[\varphi \mapsto (T, f)]\, A$. Given

$$\Gamma,\, \psi,\, i : \mathbb{I} \vdash b : B \qquad \text{and} \qquad \Gamma \vdash b_0 : B(i_0)[\psi \mapsto b(i_0)]$$

we want to compute $\mathsf{comp}^i\, B[\psi \mapsto b]\, b_0 : B(i_1)[\psi \mapsto b(i_1)]$.

5.2 The algorithm (CCHM §6.2)

Define the following auxiliary terms, with their contexts:

Name	Definition	Context	Type
a	$\mathsf{unglue}\, b$	$\Gamma, \psi, i : \mathbb{I}$	$A[\varphi \mapsto f\, b]$
a_0	$\mathsf{unglue}\, b_0$	Γ	$A(i_0)[\varphi(i_0) \mapsto f(i_0)\, b_0,\ \psi]a(i_0)$
δ	$\forall i.\, \varphi$	Γ	(the "always-on" part of φ)
a'_1	$\mathsf{comp}^i\, A[\psi \mapsto a]\, a_0$	Γ	$A(i_1)$
t'_1	$\mathsf{comp}^i\, T[\psi \mapsto b]\, b_0$	Γ, δ	$T(i_1)$
ω	$\mathsf{pres}^i\, f[\psi \mapsto b]\, b_0$	Γ, δ	path: $f(i_1)\, t'_1 \rightsquigarrow a'_1$
(t_1, α)	$\mathsf{equiv}\, f(i_1)[\delta \mapsto (t'_1, \omega),\ \psi](b(i_1),\ \mathsf{refl})\, a'_1$	$\Gamma, \varphi(i_1)$	$T(i_1)$ and correction path α
a_1	$\mathsf{comp}^j\, A(i_1)[\varphi(i_1) \mapsto \alpha_j,\ \psi]a(i_1)\, a'_1$	Γ	$A(i_1)$
b_1	$\mathsf{glue}[\varphi(i_1) \mapsto t_1]\, a_1$	Γ	$B(i_1)$ **(result)**

5.3 Step-by-step intuition

Step 1. Unglue to A. Project the partial tube b and base b_0 to A via unglue. Since A is fibrant (supports composition), we can compose in A.

Step 2. Compose in A. Compute a_1' — the "naive" A-level result at i_1.

Step 3. Compose in T (on $\delta = \forall i.\varphi$). When φ holds for all i, we also have a T-level computation t_1'.

Step 4. Apply the equiv operation. On the face $\varphi(i_1)$, use the equivalence $f(i_1)$ to find a T-element t_1 mapping (up to a path α) to a_1'. The correction path α accounts for the discrepancy between the A-composition and the image of the T-composition.

Step 5. Correct in A. Compose along α to get the final A-element a_1 that is compatible with t_1.

Step 6. Glue. Wrap up as $\mathsf{glue}[\varphi(i_1) \mapsto t_1]\, a_1$. This is well-typed because $a_1 : A(i_1)[\varphi(i_1) \mapsto f(i_1)\, t_1]$.

Remark 5.1. *When $\varphi = 1_{\mathbb{F}}$ (so $B = T$ everywhere), the algorithm reduces to $\mathsf{comp}^i\, T[\psi \mapsto b]\, b_0$, as expected from the $1_{\mathbb{F}}$-reduction.*

6 Composition for the Universe

6.1 Constructing an equivalence from a line of types

Given $\Gamma, i : \mathbb{I} \vdash E$ with $E(i_0) = A$ and $E(i_1) = B$, define:

$$
\begin{aligned}
f &= \lambda x : A.\ \mathsf{transp}^i\, E\, x && \text{(type } A \to B\text{)}\\
g &= \lambda y : B.\ (\mathsf{transp}^i\, E(i/1-i)\, y)(i/1-i) && \text{(type } B \to A\text{)}\\
u &= \lambda x : A.\ \mathsf{fill}^i\, E\, []\, x && \text{(context } i : \mathbb{I}\text{, type } A \to E\text{)}\\
v &= \lambda y : B.\ (\mathsf{fill}^i\, E(i/1-i)\, []\, y)(i/1-i) && \text{(context } i : \mathbb{I}\text{, type } B \to E\text{)}
\end{aligned}
$$

satisfying $u(i_0) = \mathrm{id}_A$, $u(i_1) = f$, $v(i_0) = g$, $v(i_1) = \mathrm{id}_B$.

One then proves f is an equivalence by constructing contractible fibers using nested fills and compositions (§7.1 of CCHM), obtaining $\mathsf{equiv}^i\, E : \mathsf{Equiv}\, A\, B$.

6.2 Composition for $\mathcal{U}$

$$\boxed{\mathsf{comp}^i\, \mathcal{U}[\varphi \mapsto E]\, A_0 \;=\; \mathsf{Glue}\big[\varphi \mapsto \big(E(i_1),\ \mathsf{equiv}^i\, E(i/1-i)\big)\big]\, A_0 \;:\; \mathcal{U}} \tag{1}$$

To compose along a partial line of types E over φ from base A_0, we construct the Glue type attaching $E(i_1)$ via the equivalence derived from the line. The reversal $i/1-i$ gives an equivalence $E(i_1) \simeq A_0$.

7 From Equivalence to Path

Example 7.1 (ua: Equiv to Path)**.** *Given $\Gamma \vdash f : \mathsf{Equiv}\, A\, B$, define in context $\Gamma, i : \mathbb{I}$:*

$$E \;=\; \mathsf{Glue}\big[(i = i_0) \mapsto (A, f),\ (i = i_1) \mapsto (B, \mathrm{id}_B)\big]\, B$$

where $\mathrm{id}_B = (\lambda x.\, x,\ \ldots) : \mathsf{Equiv}\, B\, B$. Then $E(i_0) = A$ and $E(i_1) = B$ by $1_{\mathbb{F}}$-reduction. So:

$$\mathsf{ua} \;=\; \lambda A\, B.\ \lambda f : \mathsf{Equiv}\, A\, B.\ \langle i \rangle\, \mathsf{Glue}\big[(i = i_0) \mapsto (A, f),\ (i = i_1) \mapsto (B, \mathrm{id}_B)\big]\, B \;:\; (A\, B : \mathcal{U}) \to \mathsf{Equiv}\, A\, B \to \mathsf{Pat}$$

8 The Univalence Axiom as a Theorem

8.1 The main theorem

Theorem 8.1 (Theorem 9 of CCHM)**.** *Let* $B = \mathsf{Glue}[\varphi \mapsto (T, f)]\, A$*. The map* $\mathsf{unglue} : B \to A$ *is an equivalence.*

Proof. By Lemma 2.2 (the equiv operation), it suffices to construct, given

$$\Gamma,\, \psi \vdash b : B, \qquad \Gamma \vdash u : A, \qquad \Gamma,\, \psi \vdash \alpha : \mathsf{Path}_A\, u\, (\mathsf{unglue}\, b)$$

a total lift:

$$\tilde{b} : B[\psi \mapsto b] \qquad \text{and} \qquad \tilde{\alpha} : \mathsf{Path}_A\, u\, (\mathsf{unglue}\, \tilde{b})[\psi \mapsto \alpha]$$

Step 1: Lift on face φ. On the combined face $\varphi \wedge \psi$ we have $b : T$ (since Glue elements are T-elements on φ) and $\alpha : \mathsf{Path}_A\, u\, (f\, b)$. Applying Lemma 2.2 to f (which is an equivalence on φ):

$$\Gamma,\, \varphi \vdash t : T[\psi \mapsto b] \qquad \Gamma,\, \varphi \vdash \beta : \mathsf{Path}_A\, u\, (f\, t)[\psi \mapsto \alpha]$$

Step 2: Compose in A. Define

$$\tilde{a} = \mathsf{comp}^i\, A[\varphi \mapsto \beta_i,\, \psi]\alpha_i\, u \;:\; A$$

using β on φ and α on ψ as the partial tube.

Step 3: Construct the lift. Set

$$\tilde{b} = \mathsf{glue}[\varphi \mapsto t]\, \tilde{a} \qquad \text{and} \qquad \tilde{\alpha} = \mathsf{fill}^i\, A[\varphi \mapsto \beta_i,\, \psi]\alpha_i\, u$$

Verification:

- $\tilde{b} : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A$ since $t : T$ on φ and $\tilde{a} : A[\varphi \mapsto f\, t]$. ✓
- $\tilde{b}\big|_{\psi} = \mathsf{glue}[\varphi \mapsto b]\, (\mathsf{unglue}\, b) = b$ by the η-rule. ✓
- $\mathsf{unglue}\, \tilde{b} = \mathsf{unglue}\, (\mathsf{glue}[\varphi \mapsto t]\, \tilde{a}) = \tilde{a}$ by the β-rule. ✓
- $\tilde{\alpha} : \mathsf{Path}_A\, u\, \tilde{a}$ with $\tilde{\alpha}\big|_{\psi} = \alpha$ from the fill boundary. ✓

This provides the required lift, so unglue is an equivalence. □

8.2 Contractibility and the univalence axiom

Corollary 8.2 (Corollary 10 of CCHM)**.** *For any* $A : \mathcal{U}$*, the type* $(X : \mathcal{U}) \times \mathsf{Equiv}\, X\, A$ *is contractible.*

Proof. By Lemma 5 of CCHM, it suffices to show any partial element $\varphi \vdash (T, f) : (X : \mathcal{U}) \times \mathsf{Equiv}\, X\, A$ is path-equal to the restriction of a total element. The map $\mathsf{unglue} : \mathsf{Glue}[\varphi \mapsto (T, f)]\, A \to A$ is an equivalence by Theorem 8.1. Since proofs of $\mathsf{isEquiv}\, X\, A\, f.1$ are path-equal (the type is a proposition), any partial element of $(X : \mathcal{U}) \times \mathsf{Equiv}\, X\, A$ is path-equal to the restriction of the total element $(\mathsf{Glue}[\varphi \mapsto (T, f)]\, A,\ \mathsf{unglue})$. □

Corollary 8.3 (Univalence Axiom — Corollary 11 of CCHM)**.** *For any* $A, B : \mathcal{U}$*, the canonical map*

$$\mathsf{idToEquiv} : \mathsf{Path}_{\mathcal{U}}\, A\, B \to \mathsf{Equiv}\, A\, B$$

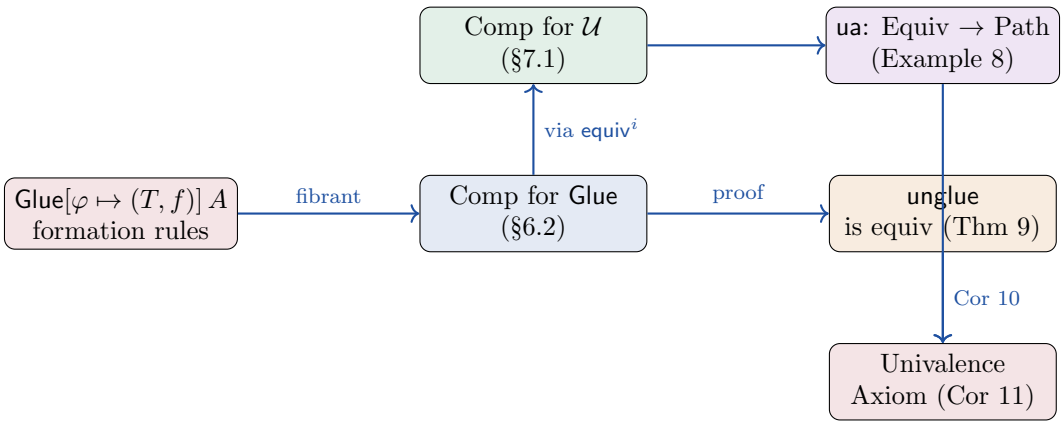
is an equivalence.

Proof. Both $(X : \mathcal{U}) \times \mathsf{Path}_{\mathcal{U}}\, A\, X$ and $(X : \mathcal{U}) \times \mathsf{Equiv}\, A\, X$ are contractible (the former by singleton contractibility; the latter by the previous corollary). A map between contractible types is an equivalence; the result follows by applying this to the fibers at B. (See Theorem 4.7.7 of the HoTT book, or Appendix B of CCHM for two alternative direct proofs.) □

Remark 8.4. *Univalence is* not *an axiom in CCHM — it is a* theorem *that follows from:*

1. *The* Glue *type former with* $1_{\mathbb{F}}$*-reduction,* η*, and* β *rules;*
2. *The composition algorithm for* Glue *(Section 6.2), making it fibrant;*
3. *Composition for the universe* $\mathcal{U}$ *(Section 7.1).*

9 Summary



Concept	**Role**	**Ref**
Glue$[\varphi \mapsto (T, f)]\, A$	Glue T onto A via equiv f on face φ	§6.1
glue$[\varphi \mapsto t]\, a$	Introduce: pair partial T-term with A-term	§6.1
unglue$[\varphi \mapsto f]\, b$	Eliminate: extract A-component, extending f	§6.1
Comp for Glue	Makes Glue fibrant; 6-step algorithm	§6.2
equiv$^i\, E$	Equivalence from a line of types E	§7.1
comp$^i\, \mathcal{U}[\varphi \mapsto E]\, A_0$	Composition for the universe	§7.1
ua	Path from equivalence	§7, Ex. 8
Thm 9	unglue is an equivalence	§7.2
Cor 11	Univalence axiom	§7.2

The central insight: Glue types are precisely the device that lets us transport fibrancy across an equivalence boundary. Once we have Glue, univalence falls out as a theorem.